Milestone 2

Group 9

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Integration and baseline generation

CLIP converts each caption into embeddings that condition Stable Diffusion 1.5's UNet. The notebook predicts and removes noise in latent space, then uses the VAE to decode an image. We used 1,000 Flickr30k image-caption pairs, split by image into 800 training, 100 validation and 100 test pairs; the test set was unused. Five pretrained baseline images used 512 x 512 pixels, 20 DDIM steps, CFG 7.5 and seeds 42-46.

Training results and early samples

On Colab GPU, we trained rank-4 LoRA adapters for 200 updates with batch size 1 and AdamW learning rate 0.0001. CLIP, VAE and original UNet weights stayed fixed. Validation compared predicted noise with the known added noise on four fixed cases. Mean squared error fell from 0.177320 to 0.177047, a 0.15% decrease. Validation ran before training and every 50 updates; blank log entries mean no validation was run at those steps.

Five LoRA outputs at step 200 on separate demonstration prompts are shown below. They match the baseline prompts, seeds and settings, giving ten saved images for visual comparison.

p01 p02 p03 p04 p05



Guidance and sampling schedule comparison

Classifier-free guidance (CFG) controls how strongly generation follows the caption. The amount of noise added at each training timestep stayed unchanged. We tuned how generation removes noise: DDIM trailing shifts the sampling timesteps toward higher noise than DDIM leading; DPM++ Karras uses a different denoising solver and changes the spacing of noise levels between steps.

We compared five settings using the pretrained model, 512 x 512 pixels, 20 steps and matched seeds. V1 describes a man dining with two women; V2 describes a yellow car on a snowy path. CLIP scores measure image-caption similarity (cosine similarity multiplied by 100); higher means closer text alignment.

Inference configuration	CFG	V1 dining	V2 car	Mean
DDIM leading baseline	7.5	30.15	34.66	32.41
DDIM leading	3.0	31.96	33.60	32.78
DDIM leading	12.0	30.38	34.82	32.60
DDIM trailing	7.5	29.54	35.47	32.51
DPM++ Karras	7.5	27.88	33.79	30.84

CFG 3 improves V1 from 30.15 to 31.96 but reduces V2 from 34.66 to 33.60. Its mean of 32.78 exceeds the baseline 32.41 because the V1 gain outweighs the V2 loss; it does not improve both prompts.

Visual observations and next steps

The paired images show no consistent benefit from the short LoRA run: the dog does not clearly jump over the log, and the child image still fails to show a correctly formed red umbrella. The small reduction in noise prediction error does not demonstrate that these visual errors were fixed. Four validation cases and two tuning captions are too few for a general conclusion. Next, train longer and compare more held-out captions and seeds using both scores and visual checks.

https://github.com/Sneha0504/IE7615_Generative_Project